

Complete System Architecture (HLD)

Education LMS for Coaching Institute (100–1000 Students)

1. System Overview

Flutter Mobile Apps (Student, Teacher, Admin) communicate securely with a NestJS backend over HTTPS.

The backend manages authentication, RBAC, courses, content, tests, attendance, notifications and analytics.

PostgreSQL stores application data, Cloud Object Storage stores media files, and Firebase Cloud Messaging delivers push notifications.

2. Architecture Layers

Presentation Layer: Flutter Apps, Admin Web.

Business Layer: Authentication, Users, Courses, Lectures, Tests, Attendance, Notifications, Analytics.

Data Layer: PostgreSQL via Prisma ORM.

Infrastructure Layer: Nginx, Docker, VPS, Cloudflare, Object Storage.

3. Client Architecture

UI Layer → State Management (Riverpod/Bloc) → Repository Layer → Network Layer (Dio) → Backend APIs.

4. Backend Modules

Authentication, User Management, Course Management, Lecture Management, Content Service, Assessment Engine, Attendance, Notifications, Analytics, Payments (optional), Chat (optional).

5. Authentication & RBAC

Login → JWT Access Token → Refresh Token → Role Lookup → Permission Check → API Authorization.

Roles: Super Admin, Admin, Teacher, Student.

6. Course Hierarchy

Course → Subject → Chapter → Lecture → Resources (Video/PDF/PPT) → Quiz.

7. Deployment

Developer → GitHub → GitHub Actions → Docker → VPS → Nginx → NestJS → PostgreSQL.
Flutter builds through Codemagic. Admin Web deployed on Vercel.

8. Security

HTTPS, JWT Authentication, RBAC, Password Hashing (Argon2/Bcrypt), Input Validation, Rate Limiting, Signed URLs for videos, Daily Backups, Audit Logs.

9. Scalability

Optimized for ~100 students on a single VPS.
Vertical scaling supports ~1000 users.
Future migration to microservices only when required.

10. Technology Stack

Flutter, Next.js, NestJS, TypeScript, Prisma ORM, PostgreSQL, Redis (optional), Cloudflare R2/AWS S3, Firebase Cloud Messaging, Docker, Nginx, GitHub Actions.